

TrES-1b

R-1897

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130806 · created 2026-08-03T13:43:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456510.7939 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.054
Transit depth (Rp/Rs)^2	6.8e-05 +/- 0.0089 [%]
Orbital Inclination (inc)	86.52 +/- 1.48
Airmass coefficient 1 (a1)	0.067 +/- 0.055
Airmass coefficient 2 (a2)	2.79 +/- 1.4
Scatter in the residuals of the lightcurve fit is	132.78 %
Best Comparison Star	#2 - [551, 426]
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.069 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.054 vs 0.1358 (deviation 2.46σ)
Duration fit vs published	0.069 d vs 0.1042 d (deviation 1.45σ)
Mid-time O–C	-0.2 min
Depth SNR	0.0
Residual scatter	132.78 %

Light curve

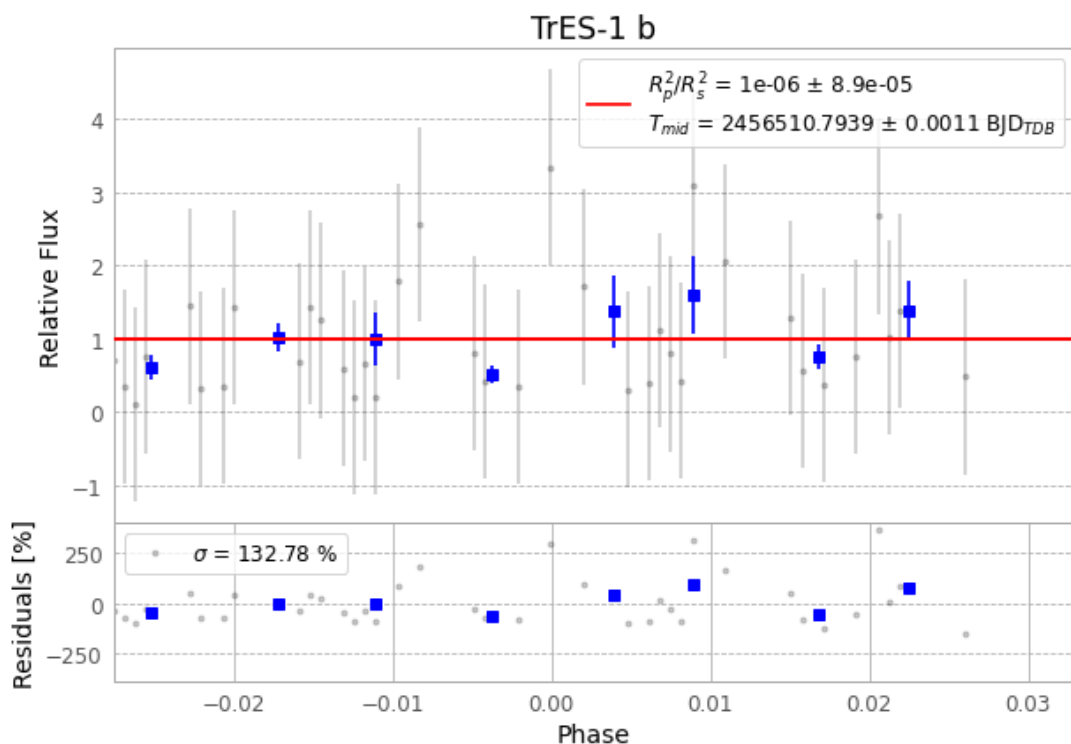


Figure 1 — FinalLightCurve_TrES-1 b_2013-08-05.png (SHA-256 fe92467b7909a90f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 414], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 879992ab5b6f97c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), TRES-1130806045712.FITS → TRES-1130806092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130806.log (e90bae031e02...), FinalLightCurve_TrES-1 b_2013-08-05.png (fe92467b7909...), FinalLightCurve_TrES-1 b_2013-08-05.csv (c402e3139080...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:43Z.